

MGT 595

Quantitative Investing

Equilibrium pricing, portfolio construction, and the empirical evidence on what actually predicts returns — built and tested with code.

Professor Tobias J. Moskowitz

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CLASS	Tuesdays, 8:30–11:30am · Classroom 4410, Evans Hall
OFFICE HOURS	Tuesdays, 1–2pm, or by appointment only
TEACHING ASST	Yiyuan Wang, Ph.D. student in Finance · yiyuan.wang@yale.edu

FORMAT

13 sessions · 3 hours each

WORKLOAD

10–20 hours per week

ASSESSMENT

11 problem sets · final exam

Course Description

This course develops and applies quantitative models for investing. Part of that process is determining the equilibrium pricing of securities in capital markets, which can then point you to where you can predict expected returns — and ultimately grow wealth. While developing portfolio theory, we will study the extensive empirical work that characterizes movements in security prices, evaluates alternative investment and asset pricing models, and attempts to test those models and interpret the implications of those tests. This is a research oriented course with practical implementation of quantitative methods for investing. It is one of the core courses in the Masters in Asset Management program, aimed at highly motivated and technically proficient students. Undergraduates who can handle that rigor and effort are welcome to enroll.

This course is designed for students who want an up-to-date treatment of quantitative investment models and the latest academic and practitioner empirical work and findings. The course is especially appropriate for students contemplating analytical finance and quantitative investment, and provides many tools and concepts that are essential for those careers. The material is covered in a rigorous analytical manner, and students must be comfortable with multivariable calculus, linear and matrix algebra, and particularly statistics and data science. The course is meant to be challenging, with the expectation that the average student spends 10–20 hours per week on the course, outside of class.

AI in this course

There will be a heavy use of AI in the course. Problem sets are given each week with extensive AI usage. The course will help you become efficient in the use of AI for building quantitative investment models. There is ample opportunity to go as deep as you wish with AI tools in the course beyond basic expectations (the 10–20 hours). The more you put into the course, the more you will get out of it.

Prerequisites

Students should be comfortable with probability, statistics, and regression analysis. Students should also feel comfortable with the concepts of risk aversion, utility functions, and budget constraints. Use of a statistical package or programming language will be vital for the course, even with AI's help. Python is highly recommended and use of Claude Code (or better) will be essential.

SUGGESTION

Upload this syllabus into an AI tool and ask it to lay out what it thinks the most useful prerequisites for the course are — then go fill the gaps it finds.

Requirements and Grading

The requirements for the course are 11 problem sets (consisting of extensive data analysis and theoretical explanation) and an in-class final exam. The problem sets are to be done in groups of 1–3 people. Class participation is also used to determine grades.

Component	Weight
Class participation	5%
11 problem sets	55%
Final exam	40%

This course should not be taken pass/fail — to get anything out of it you need to do the work, and if you're going to do that much work you might as well get a grade for it.

Class participation

Class participation is key to understanding the material in the course. Many topics and economic questions will not have a specific answer. Therefore, dialogue and debate are an important part of this course. Students should be prepared before class.

Problem sets

Problem sets will be due most weeks and are to be done in groups of 1–3 people. Problem sets consist of data analysis, including replication of studies, extensions of those studies, some conceptual questions concerning the interpretation of the analysis, and extensive use of AI to both

help you and to challenge what the AI provides. The goal is to generate a set of quantitative tools and programs throughout the course that are applied in practice and that extend your knowledge beyond what simple AI prompts can provide. In fact, you will challenge some of the output you get from AI and work to improve its answers and help you going forward. The problem sets cumulatively develop these tools by adding on to previous analysis so that extensive quantitative models are developed by the end of the course, where students will not only have built a full set of models, but will be able to digest and critique other models they encounter and improve upon them.

You may also choose to do the problem sets individually if you wish . . . but it's a lot of work.

Honor code

Students are required to adhere to the standards of conduct of the Yale SOM Honor Code and Standards of Scholarship.

Teaching Assistant and Review Sessions

- The TA for the course is **Yiyuan Wang**, Ph.D. student in Finance.
- The TA will hold a Python and AI tutorial session in the first weeks of class — students are strongly encouraged to attend (you'll thank me later).
- The TA will hold weekly office hours. The TA's objective is to help you with data and statistics questions primarily for the upcoming problem set each week, but also to go over any lingering questions.
- You should feel free to contact the TA with any and all questions, by e-mail: yiyuan.wang@yale.edu.

Readings and Materials

Texts

Required texts: **none**.

Articles

Many of the readings for the course will be posted on the course webpage with links to the articles. Some of the articles are challenging. However, I will assign certain portions of these articles that I feel are relevant to the topics discussed in class. I have also included some optional readings, which will be discussed briefly in class. Finally, you will be prompted and expected to find further readings with AI tools and dig deeper into certain topics. You are responsible for any and everything covered in class and even some material not covered in class. This will mirror your experience at a top quant investment shop — experiencing it now will not only help you land such a position, but succeed in it.

Class notes and handouts

Lecture notes are posted on the course website on Canvas. You should either take notes electronically in class or print out the lecture notes ahead of time and take notes during class on the hard

copies, so that you can follow the lecture as we discuss the various topics. Much of the material and understanding will happen in class.

AI TUTORIAL PROTOTYPE

I have developed (with the help of a former student) an **AI tutorial prototype on each lecture note** that supplements the lectures and provides tools to help you explore the topics in more detail. This prototype is a work in progress. Please provide helpful, constructive feedback on how to improve it — we will make changes in real time, both for you and for future generations of this class. To provide incentives for innovation, I will credit students who provide useful suggestions and comments that improve its content (consider it part of class participation).

Feedback, Questions, and Concerns

This course is conceptually and analytically challenging, and will require a large time commitment from students. If you have any questions about the course, please let me know. The best way to catch me is to e-mail me to set up an appointment.

Schedule of Classes

Date	Lecture topic	Due
Sept 8	Modern Portfolio Theory in Practice	
Sept 15	Evaluating Assets: Cross-Section	Problem set #1
Sept 22	Evaluating Assets: Time-Series	Problem set #2
Sept 29	Market Efficiency and Limits to Arbitrage	Problem set #3
Oct 6	Value Investing	Problem set #4
Oct 13	Momentum Investing	Problem set #5
Oct 20	<i>No class — mid-semester break</i>	
Oct 27	Quality and Defensive Investing	Problem set #6
Nov 3	How Reliable is a Backtest?	Problem set #7
Nov 10	Other Asset Markets	Problem set #8
Nov 17	Carry Strategies	
Nov 24	<i>No class — Thanksgiving break</i>	
Dec 1	Implementation Costs and Illiquidity	Problem set #9
Dec 8	Building Models and Portfolios	Problem set #10
Dec 15	Factor Timing and Crowding	Problem set #11

Course Outline

How to read this outline

- Lecture note, obtained from the course website or as a hard copy in class.
 - ★ Required reading.
 - ★ **DISCUSSED IN CLASS** Required, *and* an article we will work through and debate together in class.
- Unmarked entries are background and optional depth — read them when a topic grabs you.

Citations give the final published version where one exists. Papers still in progress are identified as working papers.

BEFORE THE COURSE BEGINS — you should know this material before entering the class.

0 Preliminaries: Background and Statistics

Not covered in class, but you **need to know** it.

- Lecture 0: Stock returns, portfolio mathematics, return distributions, and the Market Model
Fama, Eugene F., 1976, *Foundations of Finance: Portfolio Decisions and Securities Prices* (Basic Books, New York), chs. 1–4.

LECTURE TOPICS COVERED IN CLASS

1 Modern Portfolio Theory in Practice

- Course outline and introduction
- Lecture 1: Portfolio Theory, the CAPM, and Practice
- ★ Roll, Richard, 1977, “A Critique of the Asset Pricing Theory’s Tests, Part I: On Past and Potential Testability of the Theory,” *Journal of Financial Economics* 4(2), 129–176. **DISCUSSED IN CLASS**
- ★ Jorion, Philippe, 1992, “Portfolio Optimization in Practice,” *Financial Analysts Journal* 48(1), 68–74. **DISCUSSED IN CLASS**

2 Evaluating Assets: Cross-Section

PROBLEM SET #1 DUE

- Lecture 2: Cross-Sectional and Time-Series Asset Pricing Tests
- Fama, Eugene F., and James D. MacBeth, 1973, “Risk, Return, and Equilibrium: Empirical Tests,” *Journal of Political Economy* 81(3), 607–636.

3 Evaluating Assets: Time-Series

PROBLEM SET #2 DUE

- (continued) Lecture 2: Cross-Sectional and Time-Series Asset Pricing Tests
- Gibbons, Michael R., Stephen A. Ross, and Jay Shanken, 1989, “A Test of the Efficiency of a Given Portfolio,” *Econometrica* 57(5), 1121–1152.
- ★ Fama, Eugene F., and Kenneth R. French, 1992, “The Cross-Section of Expected Stock Returns,” *Journal of Finance* 47(2), 427–465. **DISCUSSED IN CLASS**

4 Market Efficiency and Limits to Arbitrage

PROBLEM SET #3 DUE

- Lecture 3: Market Efficiency / Inefficiency
- ★ Bodie, Zvi, Alex Kane, and Alan J. Marcus, 2024, *Investments*, 13th ed. (McGraw Hill, New York), ch. 12 (a good review of the issues).
- ★ Shleifer, Andrei, 2000, *Inefficient Markets: An Introduction to Behavioural Finance* (Oxford University Press, Oxford), ch. 1.
- Fama, Eugene F., 1991, “Efficient Capital Markets: II,” *Journal of Finance* 46(5), 1575–1617.
- ★ Fama, Eugene F., and Kenneth R. French, 1993, “Common Risk Factors in the Returns on Stocks and Bonds,” *Journal of Financial Economics* 33(1), 3–56. **DISCUSSED IN CLASS**
- ★ Lamont, Owen A., and Richard H. Thaler, 2003, “Can the Market Add and Subtract? Mispricing in Tech Stock Carve-Outs,” *Journal of Political Economy* 111(2), 227–268. **DISCUSSED IN CLASS**
- Kaplan, Steven N., Tobias J. Moskowitz, and Berk A. Sensoy, 2013, “The Effects of Stock Lending on Security Prices: An Experiment,” *Journal of Finance* 68(5), 1891–1936.

5 Value Investing

PROBLEM SET #4 DUE

- Lecture 4: Value
- ★ Lakonishok, Josef, Andrei Shleifer, and Robert W. Vishny, 1994, “Contrarian Investment, Extrapolation, and Risk,” *Journal of Finance* 49(5), 1541–1578. **DISCUSSED IN CLASS**

- ★ Fama, Eugene F., and Kenneth R. French, 1996, “Multifactor Explanations of Asset Pricing Anomalies,” *Journal of Finance* 51(1), 55–84. **DISCUSSED IN CLASS**
- ★ Daniel, Kent, and Sheridan Titman, 1997, “Evidence on the Characteristics of Cross-Sectional Variation in Stock Returns,” *Journal of Finance* 52(1), 1–33. **DISCUSSED IN CLASS**
 - Davis, James L., Eugene F. Fama, and Kenneth R. French, 2000, “Characteristics, Covariances, and Average Returns: 1929 to 1997,” *Journal of Finance* 55(1), 389–406.
 - Berk, Jonathan B., 2000, “Sorting Out Sorts,” *Journal of Finance* 55(1), 407–427.
 - Asness, Clifford S., and Andrea Frazzini, 2013, “The Devil in HML’s Details,” *Journal of Portfolio Management* 39(4), 49–68.
- ★ Israel, Ronen, Kristoffer Laursen, and Scott A. Richardson, 2021, “Is (Systematic) Value Investing Dead?” *Journal of Portfolio Management* 47(2), Quantitative Special Issue, 38–62.
- ★ Maloney, Thomas, and Tobias J. Moskowitz, 2021, “Value and Interest Rates: Are Rates to Blame for Value’s Torments?” *Journal of Portfolio Management* 47(6), 65–87.

6 Momentum Investing

PROBLEM SET #5 DUE

■ Lecture 5: Momentum

- ★ Jegadeesh, Narasimhan, and Sheridan Titman, 1993, “Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency,” *Journal of Finance* 48(1), 65–91. **DISCUSSED IN CLASS**
- ★ Moskowitz, Tobias J., and Mark Grinblatt, 1999, “Do Industries Explain Momentum?” *Journal of Finance* 54(4), 1249–1290. **DISCUSSED IN CLASS**
 - Grinblatt, Mark, and Tobias J. Moskowitz, 2004, “Predicting Stock Price Movements from Past Returns: The Role of Consistency and Tax-Loss Selling,” *Journal of Financial Economics* 71(3), 541–579.
 - Grundy, Bruce D., and J. Spencer Martin, 2001, “Understanding the Nature of the Risks and the Source of the Rewards to Momentum Investing,” *Review of Financial Studies* 14(1), 29–78.
 - Hong, Harrison, Terence Lim, and Jeremy C. Stein, 2000, “Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies,” *Journal of Finance* 55(1), 265–295.
 - Hvidkjaer, Soeren, 2006, “A Trade-Based Analysis of Momentum,” *Review of Financial Studies* 19(2), 457–491.
- ★ Daniel, Kent, and Tobias J. Moskowitz, 2016, “Momentum Crashes,” *Journal of Financial Economics* 122(2), 221–247. **DISCUSSED IN CLASS**
 - Kelly, Bryan T., Tobias J. Moskowitz, and Seth Pruitt, 2021, “Understanding Momentum and Reversals,” *Journal of Financial Economics* 140(3), 726–743.

7 Quality and Defensive Investing

PROBLEM SET #6 DUE

■ Lecture 6: Quality and Defensive

- ★ Novy-Marx, Robert, 2013, “The Other Side of Value: The Gross Profitability Premium,” *Journal of Financial Economics* 108(1), 1–28. **DISCUSSED IN CLASS**
 - Novy-Marx, Robert, 2013, “The Quality Dimension of Value Investing,” working paper, University of Rochester.
- ★ Asness, Clifford S., Andrea Frazzini, and Lasse Heje Pedersen, 2019, “Quality Minus Junk,” *Review of Accounting Studies* 24(1), 34–112. **DISCUSSED IN CLASS**
- ★ Frazzini, Andrea, and Lasse Heje Pedersen, 2014, “Betting Against Beta,” *Journal of Financial*

Economics 111(1), 1–25. DISCUSSED IN CLASS

Frazzini, Andrea, and Lasse Heje Pedersen, 2022, “Embedded Leverage,” *Review of Asset Pricing Studies* 12(1), 1–52.

Moskowitz, Tobias J., and Kaushik Vasudevan, 2022, “Betting Without Beta,” working paper, Yale University (earlier circulated as “What Can Betting Markets Tell Us About Investor Preferences and Beliefs? Implications for Low Risk Anomalies”).

8 How Reliable is a Backtest?**PROBLEM SET #7 DUE**

■ Lecture 7: Robustness of Anomalies, Data Mining, and Backtest Reliability

★ Harvey, Campbell R., Yan Liu, and Heqing Zhu, 2016, “... and the Cross-Section of Expected Returns,” *Review of Financial Studies* 29(1), 5–68. **DISCUSSED IN CLASS**

★ Israel, Ronen, and Tobias J. Moskowitz, 2013, “The Role of Shorting, Firm Size, and Time on Market Anomalies,” *Journal of Financial Economics* 108(2), 275–301.

★ McLean, R. David, and Jeffrey Pontiff, 2016, “Does Academic Research Destroy Stock Return Predictability?” *Journal of Finance* 71(1), 5–32. **DISCUSSED IN CLASS**

★ Hou, Kewei, Chen Xue, and Lu Zhang, 2020, “Replicating Anomalies,” *Review of Financial Studies* 33(5), 2019–2133. **DISCUSSED IN CLASS**

★ Jensen, Theis Ingerslev, Bryan T. Kelly, and Lasse Heje Pedersen, 2023, “Is There a Replication Crisis in Finance?” *Journal of Finance* 78(5), 2465–2518. **DISCUSSED IN CLASS**

★ Asness, Clifford S., Andrea Frazzini, Ronen Israel, and Tobias J. Moskowitz, 2015, “Fact, Fiction, and Momentum Investing,” *Journal of Portfolio Management* 40(5), 40th Anniversary Issue, 75–92.

★ Asness, Clifford S., Andrea Frazzini, Ronen Israel, and Tobias J. Moskowitz, 2015, “Fact, Fiction, and Value Investing,” *Journal of Portfolio Management* 42(1), 34–52.

Alquist, Ron, Ronen Israel, and Tobias J. Moskowitz, 2018, “Fact, Fiction, and the Size Effect,” *Journal of Portfolio Management* 45(1), 3–30.

Fama, Eugene F., and Kenneth R. French, 2008, “Dissecting Anomalies,” *Journal of Finance* 63(4), 1653–1678.

9 Other Asset Markets**PROBLEM SET #8 DUE**

■ Lecture 8: Other Asset Classes

★ Fama, Eugene F., and Kenneth R. French, 2012, “Size, Value, and Momentum in International Stock Returns,” *Journal of Financial Economics* 105(3), 457–472.

★ Asness, Clifford S., Tobias J. Moskowitz, and Lasse Heje Pedersen, 2013, “Value and Momentum Everywhere,” *Journal of Finance* 68(3), 929–985. **DISCUSSED IN CLASS**

★ Moskowitz, Tobias J., Yao Hua Ooi, and Lasse Heje Pedersen, 2012, “Time Series Momentum,” *Journal of Financial Economics* 104(2), 228–250. **DISCUSSED IN CLASS**

★ Moskowitz, Tobias J., 2021, “Asset Pricing and Sports Betting,” *Journal of Finance* 76(6), 3153–3209.

10 Carry Strategies

■ Lecture 9: Carry Strategies

★ Brunnermeier, Markus K., Stefan Nagel, and Lasse Heje Pedersen, 2008, “Carry Trades and Currency Crashes,” *NBER Macroeconomics Annual* 23, 313–348. **DISCUSSED IN CLASS**

- ★ Koijen, Ralph S. J., Tobias J. Moskowitz, Lasse Heje Pedersen, and Evert B. Vrugt, 2018, “Carry,” *Journal of Financial Economics* 127(2), 197–225. **DISCUSSED IN CLASS**
- ★ Lustig, Hanno, Nikolai Roussanov, and Adrien Verdelhan, 2011, “Common Risk Factors in Currency Markets,” *Review of Financial Studies* 24(11), 3731–3777.

11 Implementation Costs and Illiquidity

PROBLEM SET #9 DUE

■ Lecture 10: Trading Costs, Liquidity, and Liquidity Risk

- ★ Korajczyk, Robert A., and Ronnie Sadka, 2004, “Are Momentum Profits Robust to Trading Costs?” *Journal of Finance* 59(3), 1039–1082. **DISCUSSED IN CLASS**
- ★ Frazzini, Andrea, Ronen Israel, and Tobias J. Moskowitz, 2020, “Trading Costs,” working paper, Yale University and AQR Capital Management. **DISCUSSED IN CLASS**
- ★ Frazzini, Andrea, Ronen Israel, and Tobias J. Moskowitz, 2020, “Trading Costs of Asset Pricing Anomalies,” working paper, Yale University and AQR Capital Management. **DISCUSSED IN CLASS**
- Korajczyk, Robert A., and Ronnie Sadka, 2008, “Pricing the Commonality Across Alternative Measures of Liquidity,” *Journal of Financial Economics* 87(1), 45–72.
- ★ Amihud, Yakov, Haim Mendelson, and Lasse Heje Pedersen, 2005, “Liquidity and Asset Prices,” *Foundations and Trends in Finance* 1(4), 269–364.
- Nagel, Stefan, 2012, “Evaporating Liquidity,” *Review of Financial Studies* 25(7), 2005–2039.
- ★ Khandani, Amir E., and Andrew W. Lo, 2011, “What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data,” *Journal of Financial Markets* 14(1), 1–46.
- ★ Acharya, Viral V., and Lasse Heje Pedersen, 2005, “Asset Pricing with Liquidity Risk,” *Journal of Financial Economics* 77(2), 375–410.
- ★ Pástor, Luboš, and Robert F. Stambaugh, 2003, “Liquidity Risk and Expected Stock Returns,” *Journal of Political Economy* 111(3), 642–685.
- Novy-Marx, Robert, and Mihail Velikov, 2016, “A Taxonomy of Anomalies and Their Trading Costs,” *Review of Financial Studies* 29(1), 104–147.

12 Building Models and Portfolios

PROBLEM SET #10 DUE

■ Lecture 11: Building Models and Portfolios

- ★ Fama, Eugene F., and Kenneth R. French, 2015, “A Five-Factor Asset Pricing Model,” *Journal of Financial Economics* 116(1), 1–22. **DISCUSSED IN CLASS**
- ★ Asness, Clifford S., 2014, “Our Model Goes to Six and Saves Value from Redundancy Along the Way,” *Cliff’s Perspectives*, AQR Capital Management, December 17. **DISCUSSED IN CLASS**
- ★ Asness, Clifford S., Andrea Frazzini, Ronen Israel, Tobias J. Moskowitz, and Lasse Heje Pedersen, 2018, “Size Matters, If You Control Your Junk,” *Journal of Financial Economics* 129(3), 479–509. **DISCUSSED IN CLASS**
- ★ Asness, Clifford S., Antti Ilmanen, Ronen Israel, and Tobias J. Moskowitz, 2015, “Investing with Style,” *Journal of Investment Management* 13(1), 27–63.
- Lewellen, Jonathan, 2015, “The Cross-Section of Expected Stock Returns,” *Critical Finance Review* 4(1), 1–44.

13 Factor Timing and Crowding**PROBLEM SET #11 DUE**

■ Lecture 12: Factor Variation and Timing

- ★ Imanen, Antti, Ronen Israel, Rachel Lee, Tobias J. Moskowitz, and Ashwin Thapar, 2021, “How Do Factor Premia Vary Over Time? A Century of Evidence,” *Journal of Investment Management* 19(4), 15–57. **DISCUSSED IN CLASS**
 - ★ Alquist, Ron, Sarah Jiang, and Tobias J. Moskowitz, 2020, “Crowding,” working paper, Yale University.
 - ★ Cohen, Randolph B., Christopher Polk, and Tuomo Vuolteenaho, 2003, “The Value Spread,” *Journal of Finance* 58(2), 609–641.
 - ★ Moreira, Alan, and Tyler Muir, 2017, “Volatility-Managed Portfolios,” *Journal of Finance* 72(4), 1611–1644. **DISCUSSED IN CLASS**
 - ★ Haddad, Valentin, Serhiy Kozak, and Shrihari Santosh, 2020, “Factor Timing,” *Review of Financial Studies* 33(5), 1980–2018.
 - ★ Asness, Clifford S., Jacques A. Friedman, Robert J. Krail, and John M. Liew, 2000, “Style Timing: Value Versus Growth,” *Journal of Portfolio Management* 26(3), 50–60.
 - ★ Gupta, Tarun, and Bryan T. Kelly, 2019, “Factor Momentum Everywhere,” *Journal of Portfolio Management* 45(3), Quantitative Special Issue, 13–36.
 - ★ Arnott, Robert D., Vitali Kalesnik, and Juhani T. Linnainmaa, 2023, “Factor Momentum,” *Review of Financial Studies* 36(8), 3034–3070.
- Asness, Clifford S., Antti Imanen, and Thomas Maloney, 2017, “Market Timing: Sin a Little,” *Journal of Investment Management* 15(3), 23–40.
- Asness, Clifford S., 2016, “The Siren Song of Factor Timing, aka ‘Smart Beta Timing,’ aka ‘Style Timing,’” *Journal of Portfolio Management* 42(5), 1–6.
- Asness, Clifford S., Swati Chandra, Antti Imanen, and Ronen Israel, 2017, “Contrarian Factor Timing Is Deceptively Difficult,” *Journal of Portfolio Management* 43(5), 72–87.

14 Final Exam

In class. Cumulative, and drawn from everything we build over the semester.

ONE LAST THING

The more you put into this course, the more you will get out of it. Come prepared, argue with the readings, argue with the AI, and build tools you will still be using long after the final exam.